Subset Construction Method

- For every **non-deterministic finite state automaton** $\mathcal{A}$, an **equivalent deterministic finite state automaton** $\mathcal{A}_d$ can be constructed.

- Theorem: **subset construction**
  - Given an NFA $\mathcal{A}$ =(Q, Σ, **δ**, q_0, F)
  - There is an equivalent DFA $\mathcal{A}_d$ =(Q^d, Σ, $\boldsymbol{\delta}^d$, q_0^d, F^d) with

$$
\begin{aligned}
Q^d &= \{\bar{\delta}(\{q_0\}, w) \mid w \in \Sigma^*\} \subseteq \wp(Q) \\
q_0^d &= \hat{\varepsilon}(\{q_0\}) \\
F^d &= \{T \in Q^d \mid T \cap F \neq \emptyset\} \\
\delta^d &: Q^d \times \Sigma \to Q^d \\
\delta^d(T, a) &= \bar{\delta}(T, a) \quad \textit{for all } T \in Q^d
\end{aligned}
$$

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Subset Construction Method

- Subset construction method to convert NFA to DFA:
  - For every state in the NFA, determine all reachable states for every input symbol. (This is the function table for δ, which will be useful for the next step.)
  - A set of reachable states in the NFA constitutes a single state in the converted DFA.
  - Determine the start state of the DFA, $\hat{\varepsilon}(\{\boldsymbol{q_0}\})$ and add it to the worklist.
  - For every state q^d in the worklist and for every input symbol $\mathbf{a}$ determine $\delta^{\boldsymbol{d}}(q^d, \mathbf{a})$ (this is the same as $\bar{\delta}(q^d, \mathbf{a})$) and add it to the worklist.
  - Repeat until the worklist is empty.
  - Identify final states state. A DFA state q^d is a final state if it contains at least one final state of the NFA.

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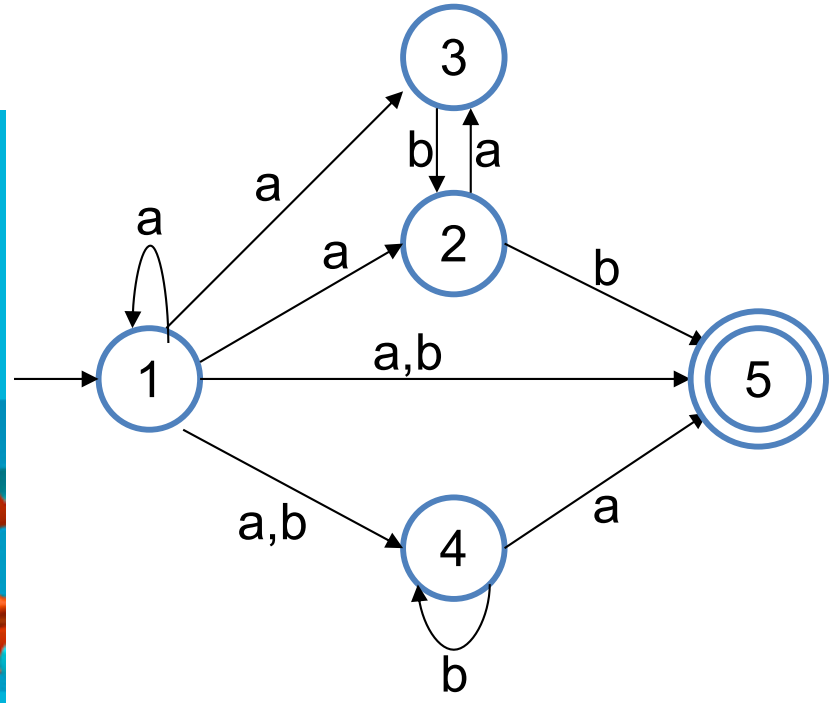
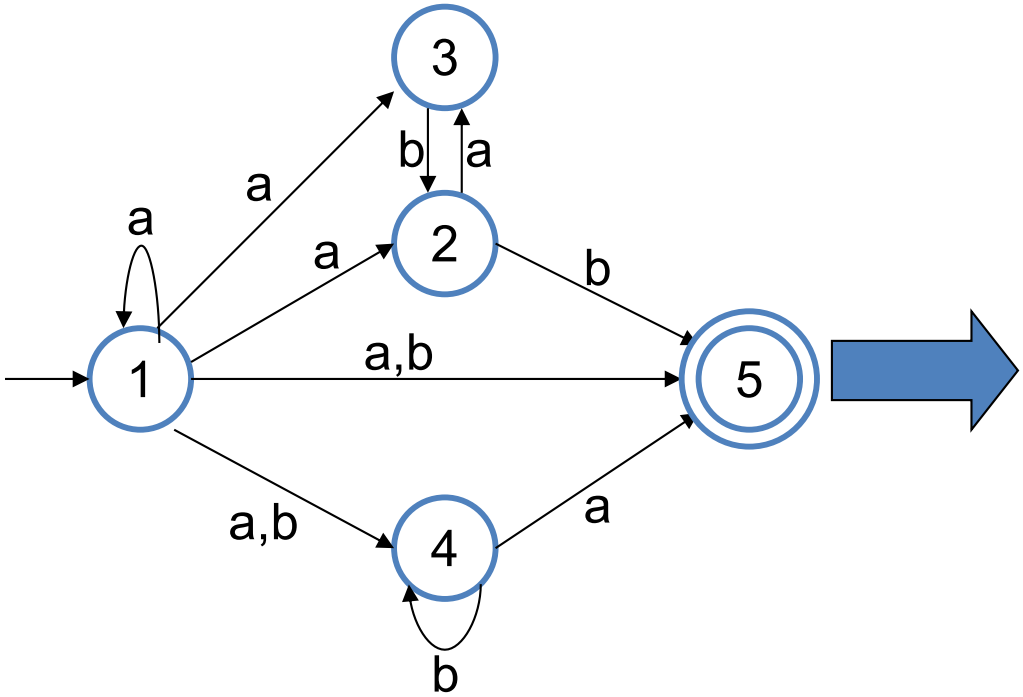
Subset Construction Method



Subset Construction Method



q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1		
2		
3		
4		
5		

Column for $\delta(q, \varepsilon)$ left out in this example, as $\delta(q, \varepsilon) = \emptyset$ for all states.

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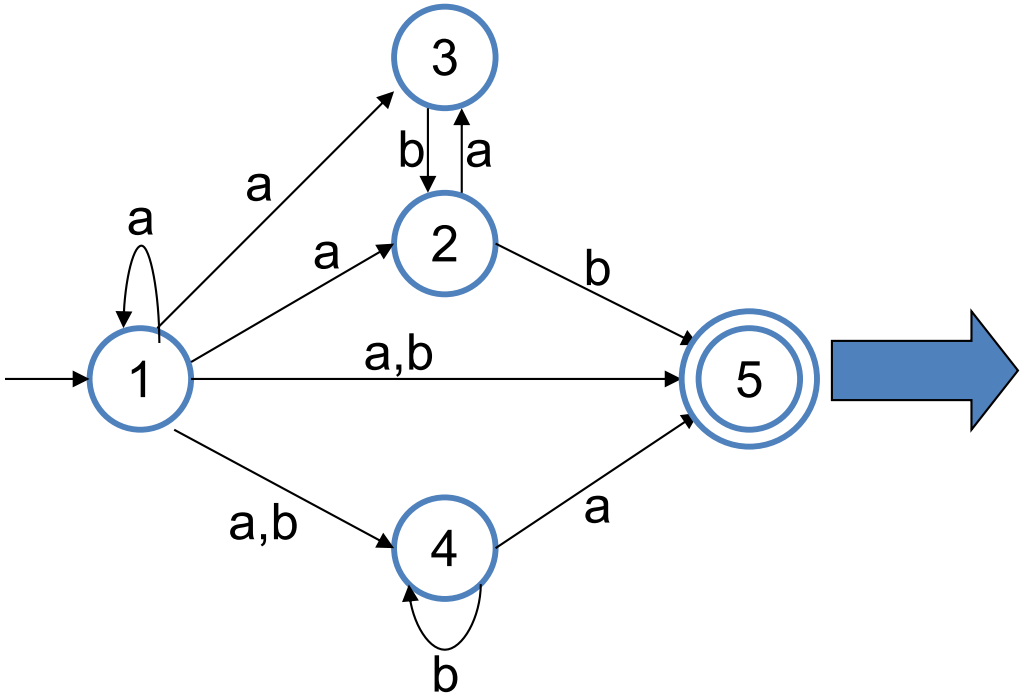
Subset Construction Method



q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Subset Construction Method

Starts with Initial state:
$\hat{\varepsilon}(\{q_0\}) = \bar{\delta}(q_0, \varepsilon) = \{1\}$

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}

There is no column for ε because epsilon transitions do not exist in a ***deterministic*** finite automaton.

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}		
{4,5}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}		
{2,4,5}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}		
{5}		
{4}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}		
{4}		
{3,5}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}		
{3,5}		
∅		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}		
∅		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅		
{2}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅	∅	∅
{2}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}		

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Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}

Subset Construction Method

q	$\delta(q, \mathrm{a})$	$\delta(q, \mathrm{b})$
1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5	∅	∅

Starts with Initial state

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}

All sets containing original state 5 (final state) are now final states.

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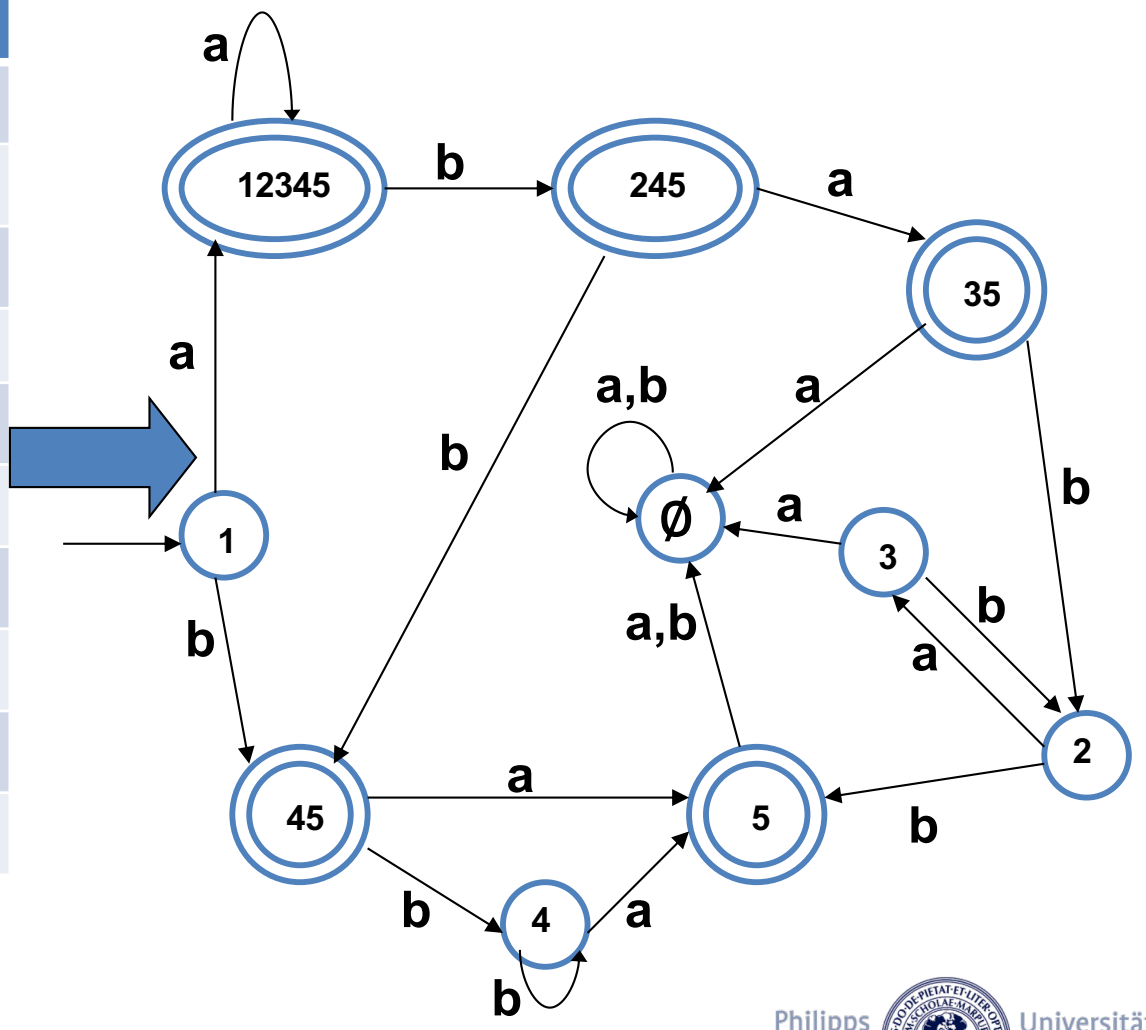
Subset Construction Method

q^d	$\delta^d(q^d, \mathrm{a})$	$\delta^d(q^d, \mathrm{b})$
{1}	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}	{1,2,3,4,5}	{2,4,5}
{4,5}	{5}	{4}
{2,4,5}	{3,5}	{4,5}
{5}	∅	∅
{4}	{5}	{4}
{3,5}	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}



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